# sysMonitor

## Service Description

Service ID: *"sysMonitor"*

**Abstract**
This document describes the native `sysMonitor` service, version 1.0.0, produced by the GraphDBManager support system of Arrowhead (documentation edition 5.3.0).

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# 1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

# 2 Overview

This document describes the `sysMonitor` service, which provides native operational telemetry of the GraphDB-Manager system: periodic `echo`, `power` and `status` observations published to the local cloud's AMQP broker. The service is produced by GraphDBManager [1]; its concrete interface is the AMQP IDD [2].

The rest of this document is organized as follows. In Section 3, we describe the abstract message operations provided by the service. In Section 4, we end the document by presenting the data types used by the mentioned operations.

## 2.1 Significant Prior Art

Arrowhead 4.x and upstream 5.2.2 had an HTTP `monitor/echo` endpoint. In this workspace the telemetry follows SysMonitor-AMQP-IDD-5.2.2; the HTTP liveness endpoint is not registered.

## 2.2 How This Service Is Meant to Be Used

The SysMonitor support system subscribes to the observations (selected with `sysMonitorManagement`) and keeps the history; LocalCloudManager shows it.

## 2.3 Important Delimitations and Dependencies

The observations describe the process, not the correctness of its answers. The service is offered only with `sysmonitor.enabled=true`. Delivery is publisher-confirmed but not end-to-end acknowledged by SysMonitor.

# 3 Service Interface

The `sysMonitor` service is a publishing service: the producer publishes one observation per operation (observation type) every publication interval (default 30 s); consumers receive them through the broker. Each operation has no input and the output Observation.

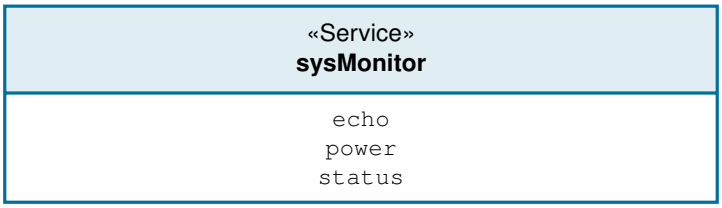



Figure 1: SysML-style block of the `sysMonitor` service and its observation types.

## 3.1 operation echo () : Observation

Liveness: `status` (`alive` or `degraded`), `uptimeSeconds` and a `detail` text; quality `good` or `uncertain`.

## 3.2 operation power () : Observation

Power supply: `supplyType=line`, `state=available`; unit `1`.

## 3.3 operation status () : Observation

Process state: `state=running`, `health` (`healthy`/`degraded`) and `serviceCount`, the number of registered services.

# 4 Information Model

## 4.1 struct Observation

One observation.

| Field | Type | Description |
|---|---|---|
| schemaVersion | Version | Version of the observation schema, `1.0`. |
| observationId | Identifier | Unique identifier of the observation. |
| providerSystem | Name | The publishing system. |
| serviceInstance | Name | The service instance (default the system name). |
| observationType | Name | `echo`, `power`, `status`. |
| sourceTime | DateTime | Time of the observation. |
| sequence | Number | Sequence number per publisher. |
| value | Metadata | The observed value (per operation). |
| unit | Name | Unit of the value, if any. |
| quality | Name | `good` or `uncertain`. |
| metadata | Metadata | `publisherMode=native`. |

## 4.2 Primitives

| Type | Description |
|---|---|
| DateTime | Pinpoints a specific moment in time (UTC). |
| Identifier | A string that identifies a record, e.g. a UUID or a composite identifier. |
| Metadata | A set of key-value pairs with arbitrary, possibly nested values. |
| Name | A string identifier that is intended to be both human and machine-readable. |
| Number | An integer number. |
| Version | Specifies a version, semantic versioning `major.minor.patch`. |

# 5 References

[1] J. Delsing, “GraphDBManager – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, ah5-GraphDBManager-java-spring/documentation/v5.3.0/GraphDBManager-v5.3.0-SysD.pdf.

[2] ——, “sysMonitor AMQP/TLS/JSON – Interface Design Description (GraphDBManager),” Eclipse Arrowhead, IDD 5.3.0, 2026, graphDBManager-v5.3.0-sysMonitor-AMQP-IDD.pdf.

# 6 Revision History

## 6.1 Amendments

| No. | Date | Version | Subject of Amendments | Author |
|---|---|---|---|---|
| 1 | 2026-10-28 | 5.2.2 | Document created from the corresponding DataManager document for the GraphDB-backed copy GraphDBManager. | Jerker Delsing |
| 2 | 2026-10-30 | 5.2.2 | Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged. | Claude Opus 5.5 (draft) |
| 3 | 2026-11-09 | 5.3.0 | New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained. | Documentation maintenance |

## 6.2 Quality Assurance

| No. | Date | Version | Approved by |
|---|---|---|---|
| 1 | | 5.3.0 | |